

1. CAUSALITY — does X actually cause Y?

Individual effect: $\tau_i = Y_i(1) - Y_i(0)$
ATE = average of τ_i over **everyone**. **ATT** = average of τ_i over **treated people only**.
Observed gap = **ATT** + **Selection bias**
Observed gap = $E[Y(1) \mid D=1] - E[Y(0) \mid D=0]$
Selection bias = $E[Y(0) \mid D=1] - E[Y(0) \mid D=0]$
In words: were the treated people already different, even without treatment?

STEPS: potential-outcomes table (Practice Q1)

- 1. New column: $\tau = Y(1) - Y(0)$ for every row.
- 2. **ATE** = average τ of all rows.
- 3. **Observed gap** = avg $Y(1)$ of treated ("Yes") rows - avg $Y(0)$ of untreated ("No") rows.
- 4. Compare: observed gap $\neq$ ATE → selection bias is present.
- 5. **ATT** = average τ of treated rows only.
- 6. **Selection** = avg $Y(0)$ of treated - avg $Y(0)$ of untreated.
- 7. **Check:** ATT + Selection = Observed gap.

- **Fundamental problem of causal inference:** we only see ONE outcome per person; the other (the **counterfactual**) is missing.
- **Random assignment (RCT):** groups alike on average → selection bias = 0 → treatment mean - control mean estimates the **ATE**. More credible because no one chooses their group.
- **Correlation $\neq$ causation:** reverse causality (Y causes X); a **confounder** drives both; chance.
- **Prediction $\neq$ policy:** X can predict Y by tracking hidden factors; changing X by policy doesn't change them.
- **Deterministic:** Y exact from X ($^{\circ}\text{F} = 1.8 \cdot ^{\circ}\text{C} + 32$).
- **Stochastic:** $Y = \beta_0 + \beta_1 X + u$ (random part u).

Data type	Who / when
Cross section	many units, ONE time (one snapshot)
Time series	one unit, many times (e.g. GDP each year)
Repeated cross sections	NEW sample each period ("fresh sample each month")
Panel	SAME units every period ("same students every fall")

Only a **panel** shows how the same person/district changes over time.

2. PROBABILITY & STATISTICS

STEPS: PMF table → probabilities, E, Var (Practice Q2)

- 1. **Valid PMF?** every $P \geq 0$ AND they add to 1.
- 2. **CDF:** running total $F(x) = P(X \leq x)$ at each value. Between values it's flat (staircase).
- 3. **"At least / more than":** 1 - rule: $P(X \geq 2) = 1 - P(X \leq 1)$.
- 4. **"Given":** $P(X \geq 2 \mid X \geq 1) = \frac{P(X \geq 2)}{P(X \geq 1)}$ (top is inside bottom).
- 5. Columns: $x \mid P \mid x \cdot P \mid x^2 \cdot P$ → add each column.
- 6. $E[X] = \text{sum of } x \cdot P$. $E[X^2] = \text{sum of } x^2 \cdot P$ (square x , THEN $\times P$).
- 7. $\text{Var} = E[X^2] - (E[X])^2 \rightarrow \text{SD} = \sqrt{\text{Var}}$. Don't square $E[X^2]$ again.
- 8. **If** $C = a + bX$ ($C = 30 + 8X$): $E[C] = a + b E[X]$; $\text{Var}(C) = b^2 \text{Var}(X)$; $\text{SD}(C) = |b| \text{SD}(X)$. Units: dollars, dollars², dollars.

1 - rule: "greater than," "at least," or "not" → 1 - (the other side). $P(X \geq 1) = 1 - P(0)$. $P(X > b) = 1 - F(b)$. $P(Z > z) = 1 - \Phi(z)$. $\Phi(-z) = 1 - \Phi(z)$.

STEPS: 2x2 joint table (Practice Q3)

- 1. Read off: $P(A)$ = its row/column **total**; $P(A \text{ and } B)$ = the **inside cell**.
- 2. $P(A \mid B) = \frac{\text{inside cell}}{B\text{'s total}}$. Name the reference group ("among rainy commutes...").
- 3. **Independence test:** compute $P(A) \cdot P(B)$, compare to the inside cell. **SAME** → **independent**. **DIFFERENT** → **NOT independent**. Write: " $P(A)P(B) = \text{---} \neq \text{---} = P(A \cap B)$, so not independent."
- 4. **0/1 variables:** $E[RL] = \text{inside cell} \rightarrow \text{Cov} = \text{cell} - P(R)P(L)$. $+ \rightarrow$ they happen together. NOT proof of causation (confounders).

STEPS: list of numbers (Practice Q4a-b)

- 1. Sort. **Mean** = sum $\div$ n . **Median** = middle value. **Mode** = most common.
- 2. Columns: $x \mid x - \bar{x} \mid (x - \bar{x})^2$ → add the last column.
- 3. $s^2 = \frac{\sum(x_i - \bar{x})^2}{n - 1}$ → **divide by $n - 1$, not n** . Then $s = \sqrt{s^2}$.
- 4. **New units** ($\times c$, e.g. min $\rightarrow$ sec, $c = 60$): mean, median, $\text{SD} \times c$; variance $\times c^2$.

$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ (+ together, - opposite).

$\text{Corr} = \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$, between -1 and 1, no units.

Corr = 0 only means no straight-line relationship. $Y = X^2$ with X centered at 0: $\text{Corr} = 0$ but Y depends completely on X .

3. THE NORMAL DISTRIBUTION

STEPS: normal probability (Practice Q4c)

- 1. Read μ and σ . $N(20, 5^2) \rightarrow \text{SD} = 5$ (**2nd number is the VARIANCE**).
- 2. z for each endpoint: $z = \frac{x - \mu}{\sigma}$ (# of SDs from mean).
- 3. Pick: $P(X \leq b) = \Phi(z_b)$; $P(X > b) = 1 - \Phi(z_b)$; between: $\Phi(z_b) - \Phi(z_a)$.
- 4. Negative z : $\Phi(-z) = 1 - \Phi(z)$. Look up Φ in the table.
- 5. Central 95% interval $\approx \mu \pm 2\sigma$ (68% = $\pm 1\sigma$, 99.7% = $\pm 3\sigma$).

z	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0	.500	.540	.579	.618	.655	.691	.726	.758	.788	.816
1	.841	.864	.885	.903	.919	.933	.945	.955	.964	.971
2	.977	.982	.986	.989	.992	.994	.995	.997	.997	.998

$\Phi(1.3)$: row "1.", column ".3" → .903. $\Phi(1.96) = .975$

Ex (between, negative z): $N(20, 5^2)$, $P(15 \leq T \leq 25)$: $z = -1$ and $+1 \rightarrow \Phi(1) - \Phi(-1) = .841 - (1 - .841) = .841 - .159 = .682$

Why $P(X = 20) = 0$: continuous → probability = **area**. A single point has no width → zero area. A range (19 to 21) has width → positive area.

4. ESTIMATORS — how good is a guess?

Estimator = the formula/rule (random). **Estimate** = the number you get from your data.

- Bias** = $E[\hat{\theta}] - \theta$. **Unbiased** = right on average. **Efficient:** among unbiased, **smaller variance**.
- Consistent:** closer to the truth as n grows (bias → 0 and variance → 0). **MSE** = $\text{Var} + \text{Bias}^2$ (lower is better).

Steps for estimator questions

- 1. Rewrite as $a + b \times (\text{base})$. Base = $\bar{X}$, X_1 , or S . (e.g. $(S + 1)/(n + 2) = 1/(n + 2) + [1/(n + 2)] \cdot S$)
- 2. Look up base: $E[\bar{X}] = \mu$, $\text{Var} = \sigma^2/n \mid E[X_1] = \mu$, $\text{Var} = \sigma^2 \mid E[S] = np$, $\text{Var} = np(1 - p)$
- 3. $E = a + b \times E[\text{base}]$
- 4. Bias = $E - \text{truth } (\mu \text{ or } p)$
- 5. $\text{Var} = b^2 \times \text{Var}[\text{base}]$ (a disappears)
- 6. **MSE** = $\text{Var} + \text{Bias}^2$. Tip: compute $p(1 - p)$ once first ($p = .52 \rightarrow .2496$)
- 7. Consistent? Huge $n \rightarrow$ bias $\rightarrow 0$ AND $\text{Var} \rightarrow 0$? Both yes = consistent.
- 8. Efficient? Only compare UNBIASED ones → smaller Var wins.

Estimator	Bias	Variance	Consistent?
$\hat{p} = S/n$	0	$p(1 - p)/n$	Yes
$\tilde{p} = \frac{S+1}{n+2}$	$\frac{1-2p}{n+2}$	$\frac{np(1-p)}{(n+2)^2}$	Yes
X_1 (for p or μ)	0	$p(1 - p)$ or σ^2	No — ignores extra data
$\bar{X}$	0	σ^2/n	Yes
$\frac{n}{n+1} \bar{X}$	$\frac{-\mu}{n+1}$	$\frac{n\sigma^2}{(n+1)^2}$	Yes

At $p = .52$, $n = 11$: $\text{MSE}(\tilde{p}) = .0227$ vs $\text{MSE}(\hat{p}) = .0163 \rightarrow \tilde{p}$ wins. MSE of $\frac{n}{n+1} \cdot \bar{X} = \frac{n\sigma^2 + \mu^2}{(n+1)^2}$.

$\bar{X}$ beats X_1 (smaller variance). Median of 0/1 data only tells you whether most answers are 1 — it doesn't estimate p .

5. REGRESSION (OLS)

Model: $Y = \beta_0 + \beta_1 X + u$. **OLS** = the line with the smallest **sum of squared residuals**.

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Also $\hat{\beta}_1 = r \cdot \frac{s_Y}{s_X} \rightarrow$ slope has the **same sign as the correlation**.

STEPS: OLS by hand from data (Practice Q7)

- 1. Find $\bar{x}$ and $\bar{y}$.
- 2. Columns: $x \mid y \mid x - \bar{x} \mid y - \bar{y} \mid (x - \bar{x})^2 \mid (x - \bar{x})(y - \bar{y})$
- 3. Add up the last two columns → $\hat{\beta}_1 = (\text{last column sum}) \div ((x - \bar{x})^2 \text{ column sum})$.
- 4. Write $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ with letters, THEN plug in numbers.
- 5. Add 2 columns: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ and $\hat{u} = y - \hat{y}$. $\hat{u} > 0 \rightarrow$ point is ABOVE the line.
- 6. **Check:** residuals add to 0, and $\sum x \hat{u} = 0$ (multiply $x \times \hat{u}$, add).
- 7. **Check:** plug $\bar{x}$ into your line, you must get $\bar{y}$. If not, the intercept is wrong.
- 8. **Predict:** $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ at the new x . Outside the data range → **extrapolation**, less reliable.

STEPS: OLS from given sums (Practice Q8c)

- 1. $\hat{\beta}_1 = \sum(x - \bar{x})(y - \bar{y}) \div \sum(x - \bar{x})^2$ (e.g. $30 \div 20 = 1.5$).
- 2. $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ (e.g. $12 - 1.5 \cdot 4 = 6$). Line: $\hat{y} = 6 + 1.5x \rightarrow$ at $x = 6$, $\hat{y} = 15$.

Changing units (Q7d, Q8d, Q9g): multiply y by $c \rightarrow$ slope and intercept both $\times c$. Multiply x by $c \rightarrow$ slope $\div c$, intercept same. (Dollars → cents: slope $\times 100$.)

Why: top of $\hat{\beta}_1$ scales by $c_x c_y$, bottom by c_x^2 ; then $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Same fitted line, just new units.

STEPS: derive OLS (Practice Q8a-b)

- 1. Minimize $Q = \sum(y_i - b_0 - b_1 x_i)^2$ (squared so + and - misses don't cancel).
- 2. $\frac{\partial Q}{\partial b_0} = -2 \sum(y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum \hat{u}_i = 0$
- 3. $\frac{\partial Q}{\partial b_1} = -2 \sum x_i(y_i - b_0 - b_1 x_i) = 0 \rightarrow \sum x_i \hat{u}_i = 0$
- 4. Divide step 2 by n : $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} \rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$; the line passes through $(\bar{x}, \bar{y})$.

Slope: "Each additional [unit of X] is associated with a $\hat{\beta}_1$ [units of Y] change in Y, on average."

Intercept: predicted Y when $X = 0$ (often meaningless if $X = 0$ is outside the data).

Not causal because: units with more X may differ in other ways (ability, income) that sit in u ; no random assignment.

Key assumption: $E[u \mid X] = 0 \rightarrow$ the "other factors" average to zero at every $X \rightarrow$ the line = **average Y at each X**. Not needed to calculate $\hat{\beta}$, only to interpret it. Fails if, e.g., income affects scores AND is related to class size. Residuals adding to 0 is just math — it does not prove $E[u \mid X] = 0$.

Reading R output (Q9f): $\widehat{\text{Score}} = 698.93 - 2.28 \cdot \text{ClassSize}$. Degrees of freedom = $n - 2 \rightarrow 418 \text{ df} \rightarrow n = 420$. R^2 (.051) = share of variation in Y explained by the line (5.1%). Descriptive only — not causal.

6. GLOSSARY — every symbol

Greek letters			
$\beta_0, \hat{\beta}_1$	true intercept, slope	$\hat{\beta}_0, \hat{\beta}_1$	estimated int., slope
μ (mu)	population mean	σ, σ^2	pop. SD, variance
ρ (rho)	pop. correlation	τ_i (tau)	i's treatment effect
$\theta, \hat{\theta}$	true value; our guess	$\Phi(z)$ (phi)	$P(Z \leq z)$ (table)
Σ (sigma)	add them all up	$\partial Q / \partial b$	derivative

Letters			
X vs. x	random var. vs. value	X_i, x_i	observation no. i
Y, X	outcome; explanatory	$Y_i(1), Y_i(0)$	if treated / not
D_i	1 = treated, 0 = not	u	error: other factors
$\hat{y}, \hat{u}$	predicted; residual	$\bar{X}, \bar{x}$	sample mean
s^2, s	sample variance, SD	s_{XY}, r_{XY}	sample cov., corr.
n	sample size	$p, \hat{p}, \tilde{p}$	true prob.; estimates
S	number of 1s in sample	Z, z	std. normal; z-score
b_0, b_1	candidate int./slope	Q	sum sq. residuals
c, a, b	constants	hat, bar	estimate, average

Functions & abbreviations		
$E[X]$	expected value (avg)	$E[Y \mid X=x]$ avg Y at that X
Var, SD	variance, std. dev.	Cov, Corr cov., correlation
$P(A \mid B)$	A given B	$A \cap B$ A and B both
PMF	$P(X = x)$ table	$F(x)$ CDF $P(X \leq x)$
$N(\mu, \sigma^2)$	normal (var. σ^2)	Bernoulli (p) 0/1, $P(1) = p$
ATE / ATT	effect: all / treated	RCT randomized trial
MSE	mean squared error	OLS ordinary least squares
df	deg. of freedom = $n - 2$	R^2 share of Y explained